

EfficientML-Cloud: A Hybrid Framework for Cost-Aware Machine Learning Inference in Enterprise Cloud Systems

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2026

Abstract

EfficientML-Cloud is a hybrid inference framework designed to reduce cloud compute cost while maintaining high predictive accuracy. The system dynamically routes inputs through either a fast lightweight model or a full deep model based on a complexity estimator and cost-sensitivity controller. This paper presents the architecture, workflow, and empirical motivation for the framework.

1 Introduction

Cloud-based machine learning inference is expensive at scale. Enterprises often deploy large models even when many inputs do not require full computation. EfficientML-Cloud addresses this by introducing a dynamic routing mechanism that selects between a fast model and a full model depending on input complexity and cost constraints.

2 System Architecture

The EfficientML-Cloud architecture consists of five major components: structured feature minimization, complexity estimation, fast and full model paths, and a cost-sensitivity controller. The goal is to reduce unnecessary computation while preserving accuracy for high-risk or high-value inputs.

3 Workflow

The workflow follows five stages:

1. **Feature Compression:** Raw enterprise data is transformed into a compact, structured feature set.
2. **Complexity Scoring:** A complexity estimator scores each input based on predicted model difficulty and risk.

3. **Routing:** Inputs are routed to either a fast model or the full model path.
4. **Cost Adjustment:** A cost-sensitivity controller adjusts routing decisions based on budget and SLA constraints.
5. **Final Inference Output:** The chosen model produces the final prediction returned to downstream systems.

4 Discussion

This hybrid approach allows enterprises to trade off cost and accuracy in a controlled manner. Low-risk, low-complexity inputs can be served by the fast model, while high-risk or complex inputs are escalated to the full model. This is particularly useful in financial services, insurance, and large-scale customer analytics.

5 Conclusion

EfficientML-Cloud provides a practical framework for cost-aware inference in cloud environments. Future work includes adaptive retraining of the fast model, multi-cloud deployment strategies, and integration with monitoring systems for real-time feedback on routing quality.

References